M13 Epinephrine (Adrenaline)

Scope of Practice	EMT • IM Auto Injector (adult and pediatric) • Prefilled Syringe (1mg/10ml) AP • Ampule (1mg/1ml)
Classification	• Adrenergic agonist

Usual Presentation	• IM EpiPen Auto-Injector:
	  • Adult: 0.3mg IM
	  • Pediatric: 0.15mg IM
	• Prefilled Syringe: 1mg/10ml
	• Ampule: 1mg/1ml
	• **Note: Presentation in mg, adrenaline infusions in mcg**

Mechanism of Action	• Adrenaline agonizes adrenergic receptors resulting in:
	  • Bronchodilation
	  • Increased heart rate and contractility.
	  • Vasoconstriction
	• Adrenaline also directly stabilizes mast cell degranulation in anaphylaxis.

Indications	• Cardiac Arrest • Shock not compatible with fluid or is fluid refractory • Bradycardia • Anaphylaxis • Life Threatening Asthma
Contraindications	• None in the emergency context
Precautions	• High risk of arrhythmias • Adrenaline will increase myocardial oxygen consumption and myocardial ischemia.
Side Effects	• Tachycardia • Arrhythmias • Tremors, anxiety and sweating • Nausea and vomiting
Adult Dose & Administration	**Cardiac Arrest:** • IV/IO: 1mg every 3-5 minutes **Septic Shock:** • IV/IO Push Dose: 0.01mg every 2-5 minutes • IV/IO Infusion: 2-10**mcg**/min **Bradycardia:** • IV/IO Push Dose: 0.01mg every 2-5 minutes • IV/IO Infusion: 2-10**mcg**/min **Anaphylaxis** • IM 0.3-0.5mg (via auto injector or 1:1,000 ampule) • IV/IO Push Dose: 0.01mg every 2-5 minutes • IV/IO Infusion: 2-10**mcg**/min **Asthma** • IM 0.3-0.5mg (via auto injector or 1:1,000 ampule) • IV/IO Push Dose: 0.01mg every 2-5 minutes • IV/IO Infusion: 2-10**mcg**/min

Pediatric Dose & Administration	**Cardiac Arrest:** • IV/IO: 0.01mg/kg every 3-5 minutes. **Septic Shock:** • IV/IO Push Dose: 0.005mg/kg • IV/IO Infusion: 0.05-0.3**mcg**/kg/min **Bradycardia:** • IV/IO 0.01mg/kg (every 3-5 minutes) **Anaphylaxis:** • IM 0.15mg-0.3mcg (via auto injector or 1:1,000 ampule) • IV/IO Push Dose: 0.005mg/kg • IV/IO Infusion: 0.05-0.3**mcg**/kg/min **Asthma:** • IM 0.15mg-0.3mg (via auto injector or 1:1,000 ampule) • IV Infusion: 0.1-1**mcg**/kg/min
Use in Pregnancy and Lactation	• If clinically indicated, adrenaline should be administered.
Additional Information	• Nil